

UNIT IV

$$17. \text{ Solve } p^2 + \left(x + y - \frac{2y}{x}\right)p + xy + \frac{y^2}{x^2} - y - \frac{y^2}{x} = 0. \quad (11)$$

Or

$$18. (a) \text{ Solve } y = xp + x(1 - p^2)^{1/2}. \quad (6)$$

$$(b) \text{ Solve } x(1 - x^2)\frac{dy}{dx} + (2x^2 - 1)y = x^3. \quad (5)$$

UNIT V

$$19. \text{ Solve by method of variation of parameter } 2\frac{d^2y}{dx^2} - \frac{dy}{dx} - 3y = 25e^{-x}. \quad (11)$$

Or

$$20. \text{ Solve the simultaneous equations } \frac{dx}{dt} + 2x - 3y = 5t, \quad \frac{dy}{dt} - 3x + 2y = 2e^{2t}. \quad (11)$$

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B.Tech. DEGREE EXAMINATION,
MAY 2022.

First Semester

MATHEMATICS - I

(Common to ALL Branches)

(From 2013 - 14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Find the curvature of the straight line.

2. What is the value of $\Gamma\left(\frac{1}{2}\right)$.

3. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

4. If $x = r\cos\theta$, $y = r\sin\theta$ find $\frac{\partial(x, y)}{\partial(r, \theta)}$.

5. Evaluate $\int_1^2 \int_1^2 \frac{dx dy}{xy}$.

6. Change of the order of integration $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx$.

7. Solve : $xp^2 - 2yp + x = 0$.

8. Solve : $a(xdy + 2ydx) = xydy$.

9. Solve : $(D^3 - 3D^2 + 4)y = 0$.

10. Find the particular integral of $(D^2 + 2D + 1)y = \sin x$.

SECTION B — (5 × 11 = 55 marks)

Answer ONE full question from each Unit.

UNIT I

11. (a) Find the radius of curvature at (c, c) on $xy = c^2$. (6)

(b) Find the equation of the evolute of the parabola $y^2 = 4ax$. (5)

Or

12. Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. (11)

UNIT II

13. (a)

If z is a function of x and y and $x = e^u + e^{-u}$, $y = e^{-u} - e^u$ show that $\frac{\partial z}{\partial u} \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$. (6)

(b) Find the extreme values of the function

$$f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x. \quad (5)$$

Or

14. (a) Expand $e^x \cos y$ in powers of x and y up to terms of third degree. (6)

(b) The temperature T at any point (x, y, z) in spaces is $T = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (5)

UNIT III

15. Change of the order of integration and hence evaluate $\int_0^{1-2-y} \int_y^{1-2-y} xy dy dx$. (11)

Or

16. Find the volume of sphere $x^2 + y^2 + z^2 = 1$ using triple integrals. (11)